Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **2.** An infinite geometric series with common ratio r converges to a finite sum if and only if |r| < 1.

____ **3.** If a function f(x) is continuous at x = c, then f(x) is necessarily differentiable at x = c.

____ **4.** The limit lim (x -> 0) [sin(x) / x] is equal to 1.

____ **5.** The derivative of the natural exponential function f(x) = e^(3x) is f'(x) = 3e^(3x).

____ **6.** A sequence {a_n} is bounded if there exist real numbers m and M such that m ≤ a_n ≤ M for all positive integers n.

II. Choose the best answer among the given alternatives.

____ **1.** What is the derivative of f(x) = 3x^4 - 5x^2 + 7 with respect to x?

A. 12x^3 - 10x **B.** 12x^3 - 10x + 7

C. 7x^3 - 10x **D.** 12x^4 - 5x

____ **7.** What is the 10th term of an arithmetic progression whose first term is a1 = 5 and common difference d = 3?

A. 29 **B.** 32 **C.** 35 **D.** 38

____ **8.** What is the sum of the first 20 terms of the arithmetic series: 2 + 5 + 8 + 11 + ...?

A. 590 **B.** 610 **C.** 630 **D.** 650

____ **9.** Find the sum of the infinite geometric series: 12 + 6 + 3 + 3/2 + ...

A. 18 **B.** 24 **C.** 30 **D.** 36

____ **10.** Evaluate the limit: lim (x -> 3) [(x² - 9) / (x - 3)]

A. 0 **B.** 3 **C.** 6 **D.** Does not exist

____ **11.** Evaluate the limit at infinity: lim (x -> ∞) [(4x³ + 2x - 5) / (2x³ - 7x² + 1)]

A. 0 **B.** 2 **C.** 4 **D.** ∞

____ **12.** What is the derivative of f(x) = 3x⁴ - 5x² + 7x - 9?

A. 12x³ - 10x + 7 **B.** 12x³ - 5x + 7

C. 7x³ - 10x + 7 **D.** 12x⁴ - 10x² + 7

____ **13.** Find the slope of the tangent line to the curve y = x³ - 2x + 4 at the point where x = 2.

A. 8 **B.** 10 **C.** 12 **D.** 14

____ **14.** What is the equation of the tangent line to the parabola y = x² at the point (2, 4)?

A. y = 4x - 4 **B.** y = 2x **C.** y = 4x + 4 **D.** y = x + 2

____ **15.** If f(x) = (2x + 1) / (x - 3), what is f'(x)?

A. -7 / (x - 3)² **B.** 7 / (x - 3)² **C.** -5 / (x - 3)² **D.** 2 / (x - 3)²

____ **16.** Find the derivative of g(x) = (3x² - 4)⁵ with respect to x.

A. 30x(3x² - 4)⁴ **B.** 5(3x² - 4)⁴ **C.** 15x(3x² - 4)⁴ **D.** 30x²(3x² - 4)⁴

____ **17.** At what critical value of x does the quadratic function f(x) = 2x² - 8x + 5 attain its minimum value?

A. x = 1 **B.** x = 2 **C.** x = 4 **D.** x = -2

____ **18.** If the position of a moving particle along a straight line is given by s(t) = 2t³ - 9t² + 12t (in meters), at what times t is the particle momentarily at rest?

A. t = 1 s and t = 2 s **B.** t = 0 s and t = 3 s

C. t = 2 s and t = 3 s **D.** t = 1.5 s only

____ **19.** Which of the following functions has a vertical asymptote at x = 2?

A. f(x) = (x - 2) / (x² - 4)
B. f(x) = 1 / (x - 2)
C. f(x) = (x² - 4) / (x - 2)
D. f(x) = sin(x - 2)

____ **20.** What is the common ratio r of a geometric sequence if the first term is a1 = 3 and the fourth term is a4 = 24?

A. 2 **B.** 3 **C.** 4 **D.** 8

____ **21.** What is the value of the limit: lim (x -> 0) [(e^(2x) - 1) / x]?

A. 0 **B.** 1 **C.** 2 **D.** e

____ **22.** If y = ln(x² + 1), what is the first derivative dy/dx?

A. 2x / (x² + 1) **B.** 1 / (x² + 1) **C.** x / (x² + 1) **D.** 2 / (x² + 1)

____ **23.** In the principle of mathematical induction, what are the two essential steps required to prove a proposition P(n) for all integers n ≥ 1?

A. Base step (prove P(1)) and Inductive step (prove P(k) => P(k+1))
B. Trial step and conclusion step
C. Derivative step and integral step
D. Contrapositive step and contradiction step

____ **24.** If f(x) = sin(2x), what is f''(x) (the second derivative)?

A. -4 sin(2x) **B.** 4 cos(2x) **C.** -2 cos(2x) **D.** -4 cos(2x)

____ **25.** Find the sum: ∑ (k=1 to 50) of k [the sum of the first 50 positive integers].

A. 1225 **B.** 1275 **C.** 1300 **D.** 2550

____ **26.** What is the normal line slope to the curve y = 3x² - 5 at the point (1, -2)?

A. 6 **B.** -6 **C.** 1/6 **D.** -1/6